Cardiovascular Critical Care – Chapter Summary

ACS • Heart Failure • Arrhythmias • Valves • Arrest

ICU / EDIC Board Review • MCQ-Oriented

1. Hemodynamic Monitoring

Parameter	Normal
CO / CI	4–8 L/min / 2.5–4 L/min/m²
CVP	2–8 mmHg
PAWP	6–12 mmHg
mPAP	10–20 mmHg
SVR	800–1200 dyn·s/cm⁵
SvO_2	65–75%

- **SVR** = 80 × (MAP − CVP) / CO
- **PVR** = (mPAP − PAWP) / CO (WU); normal < 2
- **Cardiac power output** = MAP × CO / 451 (< 0.6 W poor)
- PAWP overestimates LVEDP: mitral stenosis, high PEEP, zone 1/2
- **Echo:** VTI LVOT (normal 18–22 cm), TAPSE < 17 mm = RV dysfunction

2. Acute Coronary Syndromes

- **STEMI:** primary PCI < 90 min (< 120 from first contact); lysis within 30 min if PCI not possible
- **NSTEMI:** immediate invasive (< 2 h) if unstable / arrhythmia / refractory pain / shock; early (< 24 h) if high risk
- **DAPT:** aspirin + ticagrelor / prasugrel (clopidogrel if bleeding risk)
- **Anticoagulation:** UFH during PCI
- **Universal definition of MI:** troponin rise > 99th centile + ischemia evidence
- **Type 2 MI:** supply–demand mismatch (sepsis, anemia, tachycardia)
- O_2 only if SpO_2 < 90% (AVOID, DETO2X)

3. Mechanical Complications of MI

Complication	Timing	Clue
Papillary muscle rupture	2–7 days	Acute MR, pulmonary edema (inferior MI)
VSD	3–5 days	New holosystolic murmur, O_2 step-up in RV
Free wall rupture	3–5 days	Tamponade, PEA
LV aneurysm	Weeks	Persistent ST ↑, thrombus
RV infarct	Acute	Hypotension, clear lungs, ↑JVP

- **RV infarct:** fluids cautiously, avoid nitrates / diuretics, inotropes
- **Dressler syndrome:** pericarditis weeks after MI

4. Acute Heart Failure

Profile	Congestion	Perfusion	Rx
Warm–wet (most common)	+	OK	Diuretics, vasodilators
Cold–wet	+	↓	Inotropes ± vasopressors
Cold–dry	–	↓	Fluids cautiously
Warm–dry	–	OK	Compensated

- **Loop diuretics** IV (≥ 1–2.5 × home dose); natriuresis-guided (urine Na > 50–70 mmol/L at 2 h)
- **Acetazolamide** add-on (ADVOR); thiazide for resistance
- **Nitrates** if SBP > 110; **CPAP/NIV** for pulmonary edema
- Start GDMT early before discharge (STRONG-HF)

5. Inotropes & Mechanical Support

Agent	Mechanism	Note
Dobutamine	β_1	↑ HR, arrhythmias
Milrinone	PDE3 inhibitor	Vasodilation, renal clearance, good in β-blocked / PH
Levosimendan	Ca^{2+} sensitizer + K-ATP	Long-acting metabolite
Epinephrine	β/α	↑ lactate

- **IABP:** ↑ coronary perfusion, ↓ afterload; no mortality benefit in AMI-CS
- **Impella:** LV unloading; DanGer Shock ↓ mortality (STEMI-CS)
- **VA-ECMO:** ↑ LV afterload → LV distension → venting (Impella / IABP / septostomy)

6. Tachyarrhythmias

- **Unstable** (shock, syncope, ischemia, HF) → **synchronized cardioversion**
- **AF in ICU:** rate control (β-blocker, amiodarone, digoxin, diltiazem if EF preserved); Mg if low
- **AF anticoagulation:** CHA_2DS_2-VA ≥ 2; cardiovert < 48 h or after TOE / 3 weeks anticoagulation
- **SVT:** vagal → adenosine 6 → 12 → 18 mg
- **WPW + AF:** procainamide / amiodarone / DC shock; **avoid** AV nodal blockers
- **Monomorphic VT:** amiodarone, procainamide, DC shock
- **Torsades:** Mg 2 g IV, overdrive pacing / isoprenaline, stop QT drugs
- **Electrical storm:** ≥ 3 VT/VF in 24 h → amiodarone + β-blocker, sedation, ablation, stellate ganglion block

7. Valvular Emergencies

- **Severe AS:** avoid tachycardia & hypotension; keep preload; phenylephrine; TAVI / balloon valvuloplasty
- **Acute AR:** dissection / endocarditis → urgent surgery; avoid IABP & bradycardia
- **Acute MR:** afterload reduction, IABP, surgery
- **Mitral stenosis:** keep HR slow (β-blocker), avoid tachycardia; AF poorly tolerated
- **Dynamic LVOTO** (HOCM, takotsubo, hypovolemia): fluids, β-blocker, phenylephrine; **avoid inotropes**

8. Infective Endocarditis

- **Duke-ISCVID 2023:** major – positive cultures with typical organism, imaging (echo, PET-CT)
- **TOE** more sensitive than TTE (prosthetic, abscess)
- **Empiric:** vancomycin + gentamicin/ceftriaxone (native); + rifampicin if prosthetic
- **Urgent surgery:** HF, uncontrolled infection (abscess, fungal, resistant), vegetation > 10 mm with embolism
- Oral step-down possible after stabilization (POET)
- Stroke: surgery not delayed if no hemorrhage / coma

9. Aortic Dissection

- **Stanford A** (ascending) → emergency surgery; **B** → medical unless complicated
- **Targets:** HR < 60, SBP 100–120 – **β-blocker first** (esmolol / labetalol) then vasodilator
- CTA diagnostic; TOE if unstable
- **Complicated type B** (malperfusion, rupture, refractory pain) → TEVAR
- **Avoid:** thrombolysis, IABP

10. Pericardial Disease

- **Tamponade:** Beck's triad, pulsus paradoxus > 10 mmHg, electrical alternans
- **Echo:** RA systolic & RV diastolic collapse, plethoric IVC, respiratory variation of mitral inflow > 25%
- **Rx:** fluids temporize; **pericardiocentesis**; avoid positive pressure ventilation if possible
- **Pericarditis:** diffuse ST ↑ + PR depression → NSAID + colchicine
- **Post-cardiac surgery:** localized clot → surgical evacuation

11. Cardiac Surgery ICU

- **Low CO syndrome:** check preload, rhythm (pacing), contractility, tamponade
- **Vasoplegia:** NE, vasopressin, methylene blue, hydroxocobalamin
- **Bleeding:** > 200 mL/h → re-explore; protamine, TXA, ROTEM-guided products
- **Post-op AF:** 30%; β-blocker / amiodarone
- **Epicardial pacing wires:** AAI / DDD 80–90
- **CPR after cardiac surgery (CALS):** 3 shocks for VF before compressions; emergency resternotomy within 5 min

12. Takotsubo & Myocarditis

- **Takotsubo:** postmenopausal, stress, apical ballooning; ↑ troponin, normal coronaries
- LVOTO in 20% → avoid inotropes; levosimendan / MCS if shock
- **Myocarditis:** viral, immune checkpoint inhibitors, giant cell; CMR (Lake Louise criteria); biopsy if fulminant
- **ICI myocarditis:** high-dose steroids

13. Cardiac Arrest & Post-ROSC

- **High-quality CPR:** rate 100–120, depth 5–6 cm, minimal pauses, $EtCO_2$ > 10–20
- **Shockable:** shock → CPR 2 min; adrenaline 1 mg after 3rd shock then q3–5 min; amiodarone 300 mg (then 150 mg)
- **Non-shockable:** adrenaline 1 mg ASAP
- **4H & 4T:** hypoxia, hypovolemia, hypo/hyperkalemia, hypothermia; thrombosis (coronary / PE), tamponade, toxins, tension pneumothorax

Post-ROSC

- SpO_2 94–98%, $PaCO_2$ normal, MAP > 65
- **Temperature:** prevent fever (≤ 37.5 °C) ≥ 72 h (TTM2: 33 vs 37.5 no difference)
- Coronary angiography: urgent if STEMI; not routine without ST ↑ (COACT, TOMAHAWK)
- **Prognostication ≥ 72 h**, multimodal: absent pupillary + corneal reflexes, bilateral absent N20 (SSEP), NSE > 60 µg/L, malignant EEG, diffuse anoxic injury on CT/MRI

14. High-Yield MCQ Pearls

- ☑ RV infarct: fluids, avoid nitrates
- ☑ New murmur day 3–5 post-MI → VSD or papillary rupture
- ☑ WPW + AF: avoid AV nodal blockers
- ☑ Torsades: Mg 2 g IV
- ☑ Dynamic LVOTO: fluids + β-blocker, no inotropes
- ☑ Dissection: β-blocker before vasodilator; HR < 60
- ☑ Tamponade: avoid positive pressure ventilation
- ☑ Post-cardiac surgery VF: 3 stacked shocks
- ☑ Milrinone: renal clearance, vasodilation
- ☑ VA-ECMO ↑ LV afterload → vent
- ☑ TTM2: no benefit of 33 °C – prevent fever
- ☑ Neuroprognosis ≥ 72 h, multimodal

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